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12 April 1979
SUPERSEDING
MIL-C-49059
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MILITARY SPECIFICATION

CABLE, ELECTRICAL (FLEXIBLE, FLAT, UNSHIELDED), (FLAT CONDUCTOR) GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flat multiconductor, unshielded flexible cable suitable for use in aerospace, ground and shipboard applications where voltages do not exceed 300 volts (rms), to provide minimum size and weight.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-502	-	Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips and Bars).
QQ-C-576	-	Copper Flat Products with Slit, Slit and Edge-rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet and Strip).
TT-M-261	-	Methyl Ethyl Ketone, Technical.
TT-I-735	-	Isopropyl Alcohol.
UU-T-450	-	Tissue, Facial.

MILITARY

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-12000	-	Cable, Cord and Wire, Electric, Packaging of.
MIL-F-14256	-	Flux, Soldering Liquid (Rosin Base).
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.
MIL-C-45662	-	Calibration System Requirements.
MIL-C-49059/1	-	Cable, Electrical (Flexible, Flat, Unshielded), (Flat Conductor - Unplated), 300 Volts, 105°C.
MIL-C-49059/2	-	Cable, Electrical (Flexible, Flat, Unshielded), (Flat Conductor), 300 Volts, 150°C
MIL-C-49059/3	-	Cable, Electrical (Flexible, Flat, Unshielded), (Flat Conductor), 300 Volts, 200°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications Research and Development Command, ATTN: DRDCO-CM-DM, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-228 - Cable and Wire, Insulated; Methods of Training.

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(See Supplement 1 for list of applicable specification sheets.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B33 - Tinned Soft or Annealed Copper Wire for Electrical Purposes.
ASTM B298 - Silver-coated Soft or Annealed Copper Wire.
ASTM D3032 - Hookup Wire Insulation, Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Detail requirements for individual cable styles. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

3.2 Qualification. Cables furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Materials. Unless otherwise specified (see 3.1), the materials for the principal components of the cable shall be as specified herein. Prior approval to use substitute material must be obtained from the qualifying activity. When a definite material is not specified, a material shall be used that will enable the finished products to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Fungus resistance. Finishes and materials shall be certified that they meet the requirements of MIL-STD-454, requirement 4. Materials or finishes not meeting this certification shall be tested in accordance with method 508 of MIL-STD-810. There shall be no evidence of fungus growth.

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3.3.2 Conductors. Unless otherwise specified, conductor material shall be copper of 100 percent conductivity conforming to the following:

Slit-conductors	- Solid, soft, or annealed ETP flat copper, edge rolled, without burrs, per QQ-C-576.
Flattened-round wire conductors	- Solid, soft or drawn and annealed, bare round copper wire flattened to size.
Flat conductors	- Soft temper, oxygen free copper wire in accordance with QQ-C-502.

3.3.3 Conductor coating. Coated conductors shall be as specified in the applicable specification sheet (see 3.1) and as indicated below.

3.3.3.1 Silver-coated conductors. Silver-coated round conductors shall have a coating thickness of not less than 40 microinches of silver when tested in accordance with 4.6.1.1.1. For flat conductors constructed from round conductors, the coating requirements for round conductors shall apply prior to the flattening process.

3.3.3.2 Tin-coated conductors. Tin coated conductors shall pass the continuity and adherence tests when tested in accordance with 4.6.1.1.3.

3.3.3.3 Cleaning. Before application of the insulation, the conductor should be thoroughly cleaned to remove foreign material which may contribute to failure.

3.3.4 Insulation material. Unless otherwise specified, insulation materials shall be certified virgin material (see 3.3.4.1), containing no additives except those required as lubricants used in extrusion, pigmentation for colors or for fire or flame retardance. Insulation material shall be as specified in table I, and the applicable specification sheet (see 3.1).

TABLE I. Insulation material.

Temperature rating °C	Insulation material
105	Polyethylene Terephtalate
150	Crosslinked modified ethylene tetrafluroethylene
200	Tetrafluroethylene
200	Fluorinated ethylene propylene

3.3.4.1 Virgin material. For purposes of this specification, virgin material shall be 100 percent new material which has been through only the process essential to its manufacture and its application to the finished cable and has been through these essential processes one time only. Any material which has previously been processed in any other manner is considered nonvirgin material. This requirement shall apply to the manufacture of all ingredients and components used.

3.3.4.2 Uniformity of insulation thickness. The insulation thickness on each side of the conductors shall be as specified (see 3.1).

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3.4 Design and construction. Design and construction of the cable shall be as specified herein and in the applicable specification sheet (see 3.1).

3.4.1 Conductors. The spacing, size, and number of the conductors shall be in accordance with the applicable specification sheet (see 3.1).

3.4.2 Finished cable. Finished cable shall conform to the requirements of the applicable specification sheet (see 3.1).

3.4.2.1 Splices. All delivered cable shall be free of conductor splices. There shall be no splices of the cable as a whole unit.

3.4.2.2 Cable weight. When tested in accordance with 4.6.1.2, the maximum cable weight per thousand feet shall be in accordance with the applicable specification sheet (see 3.1 and 4.6.1.2).

3.4.2.3 Color. Unless otherwise specified, the color shall be the natural color as supplied by the manufacturer. When solid color(s) are required, these colors shall be in accordance with MIL-STD-104, class 1 (see 6.2).

3.5 Electrical requirements.

3.5.1 Insulation flaws. One hundred percent of the finished cable shall pass the insulation flaws test specified in 4.6.2.1. Test voltages shall be as specified in the applicable specification sheet (see 3.1).

3.5.2 Dielectric withstanding voltage. When tested in accordance with 4.6.2.2, the cable shall withstand the voltage specified in the applicable specification sheet (see 3.1) and shall show no evidence of breakdown or flashover.

3.5.3 Insulation resistance. When tested in accordance with 4.6.2.3, the insulation resistance shall be not less than 500 megohms - 1,000 feet.

3.5.4 Conductor resistance. When tested in accordance with 4.6.2.4, the conductor direct current (dc) resistance shall not exceed the value specified in the applicable specification sheet (see 3.1).

3.6 Conductor continuity. One hundred percent of the finished cable shall pass the conductor continuity test specified in 4.6.3. There shall be no indication of discontinuity in any conductor.

3.7 Flammability. Cables shall be tested in accordance with 4.6.4. The afterburn shall be 3 seconds (maximum), the burn length shall be 3 inches (maximum) and there shall be no flaming of facial tissue.

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3.8 Folding and folding temperature test.

3.8.1 Folding. When tested in accordance with 4.6.5, there shall be no loss of continuity in the conductors and no cracking or separation (delamanition) of the insulation. The cable shall then be subjected to the dielectric withstanding voltage requirements specified in 3.5.2.

3.8.2 Folding temperature test. When tested in accordance with 4.6.6, there shall be no loss of continuity in the conductors and no cracking or separation (delamanition) of the insulation. The cable shall then be subjected to the dielectric withstanding voltage requirements specified in 3.5.2.

3.9 High-temperature aging. When tested in accordance with 4.6.7, there shall be no cracking or separation of the insulation. The cable shall then be subjected to the dielectric withstanding voltage requirements specified in 3.5.2.

3.10 Flexing endurance. When tested in accordance with 4.6.8, there shall be no loss of continuity in the conductors and no cracking or separation (delamanition) of the insulation.

3.11 Moisture resistance. When tested in accordance with 4.6.9, the cable shall meet the insulation resistance requirement of 3.5.3. There shall be no cracking or separation of insulation.

3.12 Low pressure (maximum rated temperature). When tested in accordance with 4.6.10, the cable insulation shall show no evidence of shrinkage greater than 1/16 inch at either end of the cable specimen, ballooning, bubbling, separation of the insulation or voids when subjected to low pressure simulation at the elevated temperature specified in 4.6.10. Following this exposure, the cable shall meet the dielectric withstanding voltage and insulation resistance requirements specified in 3.5.2 and 3.5.3.

3.13 Vacuum stability. When tested in accordance with 4.6.11, the total weight loss (TWL) shall not exceed 1 percent and the volatile condensable materials (VCM) shall not exceed 0.1 percent.

3.14 Fluid immersion. When a cable is immersed in fluids in accordance with 4.6.12, there shall be no increase in cable weight greater than five percent, no loss of continuity in the conductor and no cracking or separation of the insulation. The cable shall meet the dielectric withstanding voltage requirements specified in 3.5.2.

3.15 Blocking. When cable is tested in accordance with 4.6.13, adjacent layers of the cable shall not block.

3.16 Notch sensitivity (edge). When cable is tested in accordance with 4.6.14, the tensile load required to cause fracture shall not be less than 25 pounds and the notch in the insulation shall not propagate beyond the second conductor before fracture of the first conductor.

3.17 Twist. When cable is tested in accordance with 4.6.15, there shall be no cracking or separation of insulation.

3.18 Shrinkage. When tested in accordance with 4.6.16, there shall be no separation of the insulation and the insulation shall not shrink more than 0.062 inch in 12 inches.

3.19 Strippability. The completed cable shall be capable of having the insulation removed by stripping as specified in 4.6.17. Where the insulation is stripped, there shall be no evidence of conductor damage and only a trace of insulation or adhesive remaining which can be easily removed by peeling.

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3.20 Solderability. Conductors shall withstand the test specified in 4.6.18.

3.21 Marking. Unless otherwise specified (see 3.1), cable shall be marked in accordance with MIL-STD-130 and the following:

a. Military part number (see 3.1).
b. Manufacturers name and symbol and the date and source code shall be in accordance with MIL-STD-1285.

Unless otherwise specified, the printing shall be a contrasting color in accordance with MIL-STD-104 class 1. The location of the printed marking shall conform to figure 1 and shall be at intervals of not greater than twelve inches between markings.

3.22 Durability of identification. When tested in accordance with 4.6.19, the identification marking shall withstand a minimum of 125 cycles (250 strokes).

3.23 Workmanship. The cable shall be constructed in a thoroughly workmanlike manner in accordance with accepted high-grade production techniques. The cable shall be free of kinks, abrasions, cracked, or peeled surfaces. The cable shall be a uniform and consistent product and shall be free from defects which will adversely affect the serviceability of the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

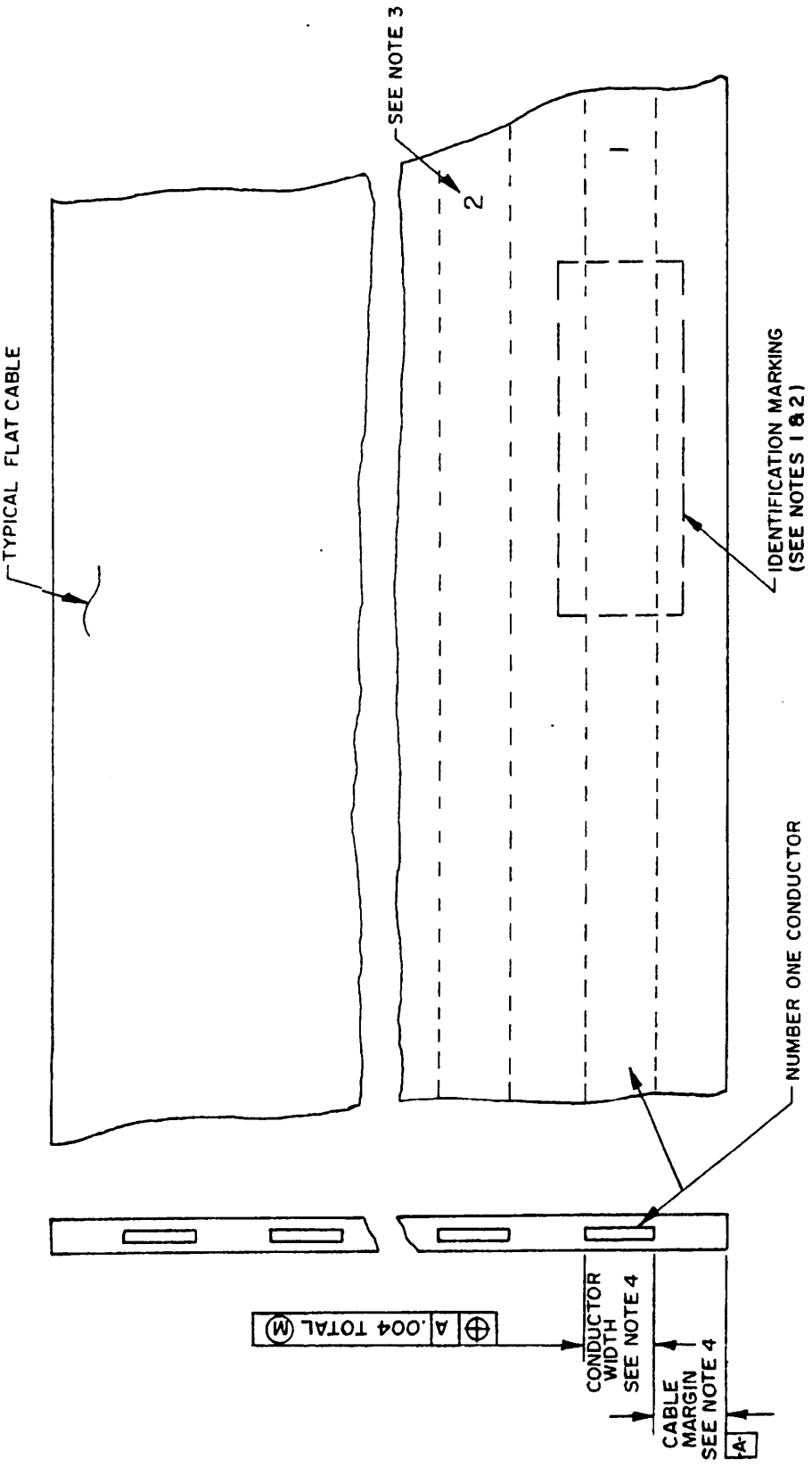
4.2 Classification of inspection. The inspections of cables are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3), on sample units produced with equipment and procedures normally used in production.

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NOTES:
1. The vertical axis of the identification marking characters shall be parallel to the longitudinal axis of the cable.
2. Identification marking shall be placed over the number one conductor, and shall not cover more than one conductor.
3. Individual conductor identification optional.
4. Conductor width and cable margin as specified in 3.1.

FIGURE 1. Identification marking.

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4.4.1 Sample size. A sample consisting of two specimens of finished cable shall be submitted to the qualification tests of table II. Each specimen shall be 20 linear feet of continuous unspliced cable. A separate sample for each specification sheet of the cable with the largest number of conductors shall be submitted. Qualification shall extend to all cables of the same specification sheet with less conductors and the same material.

4.4.2 Test routine. All samples for qualification shall have previously passed the insulation flaws test (see 4.6.2.1). The two cable lengths shall be submitted to the visual and mechanical inspections of group I in the order shown. Each of the cable lengths shall then be divided into test lengths specified for each test, and subjected to the rest of the qualification tests. One test length from each cable length shall be submitted to each test.

4.4.3 Failure. Failure of any one of the tests in table II shall be cause for refusal to grant qualification.

4.4.4 Retention of qualification. To retain qualification, the contractor shall forward a report at 12-or 36-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery, group A, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
b. The results of tests performed for qualification verification inspection, group B, including the number and mode of failures. The report shall include results of all qualification verification inspection tests performed and completed during the 36-month period. If the test results indicate nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each reporting period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative cable of those covered by a single specification sheet to testing in accordance with the qualification inspection requirements.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all cable covered by a single specification sheet (see 3.1), produced under essentially the same conditions and submitted for inspection at one time.

4.5.1.1 Unit of product. The unit of product shall be each reel, spool, or coil of cable.

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TABLE II. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical		
Conductor size	3.4.1	4.6.1
Cable dimensions and construction	3.4.2	4.6.1
Cable color	3.4.2.3	4.6.1
Marking	3.21	4.6.1
Cable weight	3.4.2.2	4.6.1
Workmanship	3.23	4.6.1
Electrical		
Insulation flaws	3.5.1	4.6.2.1
Dielectric withstanding voltage	3.5.2	4.6.2.2
Insulation resistance	3.5.3	4.6.2.3
Conductor resistance	3.5.4	4.6.2.4
Conductor continuity	3.6	4.6.3
Group II		
Flammability	3.7	4.6.4
Folding	3.8.1	4.6.5
Folding temperature	3.8.2	4.6.6
High-temperature aging	3.9	4.6.7
Twist	3.17	4.6.15
Shrinkage	3.18	4.6.16
Durability of identification	3.22	4.6.19
Group III		
Silver-coated conductors	3.3.3.1	4.6.1.1.1
Tin-coated conductors	3.3.3.2	4.6.1.1.2
Group IV		
Flexing endurance	3.10	4.6.8
Moisture resistance	3.11	4.6.9
Low pressure (maximum rated temperature)	3.12	4.6.10
Vacuum stability	3.13	4.6.11
Blocking	3.15	4.6.13
Notch sensitivity (edge)	3.16	4.6.14
Strippability	3.19	4.6.17
Solderability	3.20	4.6.18
Group V		
Fungus resistance	3.3.1	---
Fluid immersion	3.14	4.6.12

4.5.1.1.2 Sample. The sample shall consist of that number of randomly selected units of product required by the applicable sampling plan in MIL-STD-105.

4.5.1.1.3 Specimens. Specimens of inspection shall be taken from each unit of product which forms part of the sample. A specimen shall be a length of cable drawn from a unit of product.

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4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections of table III, subgroups 1, 2, and 3. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.1 Subgroup 1. For subgroup 1, the acceptable quality level (AQL) shall be as specified in table III. Subgroup 1 tests may be performed in any order.

4.5.1.2.2 Subgroup 2. For subgroup 2, the sample shall be 100 percent of the cable in the inspection lot and every length of cable shall be subjected to the test. Any portion of the cable exhibiting breakdown or nonconformity when subjected to the insulation flaws test shall be cut out and removed.

4.5.1.2.3 Subgroup 3. For subgroup 3, the AQL shall be as specified in table III. Subgroup 3 tests shall be performed in the order shown in table III.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Subgroup 1				
Visual and mechanical				
Conductor size	3.4.1	4.6.1	1.0%	4.0%
Cable dimensions and construction	3.4.2	4.6.1		
Cable color	3.4.2.3	4.6.1		
Marking	3.21	4.6.1		
Cable weight	3.4.2.2	4.6.1.2		
Conductor identification	Figure 1	4.6.1		
Workmanship	3.23	4.6.1		
Subgroup 2				
Insulation flaws	3.5.1	4.6.2.1	100% (see 4.5.1.2.2)	
Conductor continuity	3.6	4.6.3		
Subgroup 3				
Insulation resistance	3.5.3	4.6.2.3	2.5%	1/
Conductor resistance	3.5.4	4.6.2.4		
Flammability	3.7	4.6.4		
Dielectric withstanding voltage	3.5.2	4.6.2		
Folding	3.8.1	4.6.5		
Dielectric withstanding voltage	3.5.2	4.6.2		
Folding temperature test	3.8.2	4.6.6		
Dielectric withstanding voltage	3.5.2	4.6.2	1%	4.0%
Twist	3.17	4.6.15		
Dielectric withstanding voltage	3.5.2	4.6.2		
High temperature aging 105°C, 150°C and 200°C rating	3.9	4.6.7.1		
Dielectric withstanding voltage	3.5.2	4.6.2		
Shrinkage	3.18	4.6.16		
Durability of identification	3.22	4.6.19		

1/ All electrical defects are considered major.

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4.5.1.2.4 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tenured for acceptance unless the former rejection or requirement or correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.2.5 Disposition of sample units. Sample units which have passed all the group A inspection may be delivered on the contract if the lot is accepted.

4.5.2 Qualification verification inspection. Qualification verification inspection shall consist of group B inspection. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.4), delivery of products which have passed group A inspection shall not be delayed pending the results of these qualification verification inspections.

4.5.2.1 Group B inspection. Group B inspection shall consist of the inspections specified in table IV, in the order shown. Group B inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Silver-coated conductors	3.3.3.1	4.6.1.1.1
Tin-coated conductors	3.3.3.2	4.6.1.1.2
High-temperature aging	3.9	4.6.7
Flexing endurance	3.10	4.6.8
Moisture resistance	3.11	4.6.9
Fluid immersion	3.14	4.6.12
Vacuum stability	3.13	4.6.11
Blocking	3.15	4.6.13
Strippability	3.19	4.6.17
Solderability	3.20	4.6.18

4.5.2.1.1 Sampling plan. On a 3 year basis starting from the date of qualification, specimens shall be selected from the cable with the largest number of conductors that is covered by a single specification sheet and is in production. Specimens shall be the length specified in the applicable test method paragraph. One test length shall be submitted to each test.

4.5.2.1.2 Failures. If one or more specimens fail to pass group B inspection, the sample shall be considered to have failed.

4.5.2.1.3 Disposition of sample units. Sample units from which a specimen has failed any of the group B inspection shall not be delivered on the contract, even though the inspection lot submitted is accepted.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection

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shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of MIL-C-12000.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. The cable shall be examined to verify that the design, construction, physical characteristics and dimensions, color, marking, and workmanship are in accordance with the applicable requirements (see 3.4, 3.21 and 3.23).

4.6.1.1 Conductor coatings.

4.6.1.1.1 Silver-coated conductors (see 3.3.3.1). Silver-coated conductors shall be tested for continuity of coating in accordance with ASTM B298. Specimen length shall be 6 ±0.5 inches.

4.6.1.1.2 Tin-coated conductors (see 3.3.3.2). Tin-coated conductors shall be tested for continuity and adherence of coating in accordance with ASTM B33. Specimen length shall be 6 ±0.5 inches for continuity and 12 ±0.5 inches for adherence.

4.6.1.2 Cable weight (see 3.4.2.2). A specimen of cable 26.44 ±0.02 inches shall be cut and weighed in grams to the nearest 1 percent of its weight. The grams of the specimen shall be recorded as the weight in pounds per 1,000 feet.

4.6.2 Electrical tests. It shall be permissible to wind finished cable on testing spools and reels for electrical tests, and to rewind on shipping spools and reels after completion of the last test.

4.6.2.1 Insulation flaws (see 3.5.1). All cables shall be subjected to the insulation flaws test in accordance with method 6211 of FED-STD-228 with the following exceptions and conditions: The cable conductors shall be electrically tied together and grounded. The speed of the cable through the brushes shall be such that the insulation is subjected to at least 12 complete voltage cycles at the specified AC voltage or the specified DC voltage for not less than 0.2 second. A suitable fault signaling device for detection of insulation flaws shall be provided. Any flaws detected shall be removed.

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4.6.2.2 Dielectric withstanding voltage (see 3.5.2). The cable shall be tested in accordance with ASTM D 3032, section 7, with the following exceptions: The specimen shall be prepared for high-voltage connection by isolating adjacent conductor terminals from each other. This may be accomplished by carefully cutting into the ends of the specimen 1/2 inch and 1-1/2 inches, respectively, so that adjacent conductors can be bent to opposite sides of the specimen. Remove 1/4-inch of insulation from the 1-1/2 inch end for electrical connections. For the first part of the test, the voltage shall be applied between all conductors electrically connected together, and the solution. For the second part of the test, the voltage shall be applied between each conductor and its adjacent conductors. This may be accomplished by bussing together all conductors bent to one side of the specimen and connecting one high-voltage lead. The conductors bent to the other side shall be treated in a similar manner and connected to the other high-voltage lead. The specimen shall be immersed, except for 6 inches at each end, for 1 hour. Test will be run before and after each test specifying dielectric withstanding voltage.

4.6.2.3 Insulation resistance (see 3.5.3). The cable shall be tested in accordance with ASTM D 3032, section 5, using a specimen of sufficient length to allow immersion of at least 50 conductor feet. The conductors shall be electrically connected together as specified in 4.6.2.2. The insulation resistance shall be measured between all conductors connected together, and the solution. The specimen then shall be removed from the solution and the insulation resistance shall be measured between each conductor and its adjacent conductors. The insulation resistance shall be calculated as follows:

$$\text{Insulation resistance, ohms-1000 feet} = \frac{RL}{1000} \quad (K)$$

where: R = measured resistance, ohms
L = immersed length (total length when measured dry), feet
n = number of conductors in the cable
K = n when measurement is made between all conductors electrically connected together and the solution
= $\frac{n}{2}$ when measurement is made between each conductor and its adjacent conductors

This test may be performed concurrently with the voltage withstand test, after application of the voltage specified for the voltage withstand test.

4.6.2.4 Conductor resistance (see 3.5.4). The DC resistance of the individual conductors of each specimen 38 to 40 inches long, shall be tested in accordance with method 6021 of FED-STD-228, except that the cable shall not be immersed, and calculated for conformance to the applicable specification sheet (see 3.1).

4.6.3 Conductor continuity (see 3.6). All cable shall be tested for conductor continuity with an ohmmeter or other suitable testing device.

4.6.4 Flammability (see 3.7). The test shall be performed in a sheet metal cabinet conforming to FED-STD-191, Method 5903. The test specimen shall be 12 in. ±0.5 inch long and 0.50 in. ±.06 inch wide for 22 AWG equivalent and smaller, and 12 inches long and 1.00 ±.06 inch wide for 20 AWG equivalent and larger. Two in. ±.25 inch of insulation shall be removed from one end of the specimen. Three centrally located conductors shall be twisted together for mounting as shown in figure 2. Any remaining bared conductors shall be cut off at the edge of the insulation. The other end of the specimen shall be attached to a wire by means of a clamp. The wire shall pass over a pulley and have a weight attached to it as shown in figure 2. The weight shall be sufficient to hold the specimen tautly throughout the test. A 1.5-in. ±0.1 inch-high yellow flame from a Bunsen burner conforming to FED-STD-191, Method 5903, shall be applied to the specimen at the junction of the insulation and the bare conductors in such a manner that the lower end of the insulation is located 0.75 in. ±.25 inch into the flame. The burner shall be supplied with commercial natural gas. After 12 seconds $^{+1}_{-0}$ seconds of flame application, the burner shall be removed from below the specimen. The turn length, the time of burning after removal of the flame, and the presence of dripping or falling

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flaming particles shall be recorded. The burn length shall be the distance from the original bend made in the conductors to the farthest point of damage. Damage shall consist of bare conductor and charred insulation. Light discoloration of the insulation shall not constitute failure.

4.6.4.1 Apparatus. The test shall be performed within a test chamber approximately 1 foot square by 2 feet in height, open at top and front to provide adequate ventilation for combustion but to prevent drafts. The specimen holder shall be designed so that the lower end of a 24-inch cable specimen is held by a clamp, while the upper end of the specimen passes over a pulley and can be suitably weighted to hold the specimen taut at an angle of 60 degrees with the horizontal, in a plane parallel to and approximately 6 inches from the back of the chamber. The separation between clamp and pully shall be such that a minimum of 20 inches of the specimen is freely suspended. The test flame shall originate from a Bunsen type gas burner with a 1/4-inch inlet, a needle valve in the base for gas adjustment, 3/8-inch bore nominal, and an approximate 4-inch barrel length above the air inlets. The burner shall be adjusted to furnish a 3-inch conical flame with an approximate 1-inch long inner core and a flame temperature not less than 954°C (1749°F) at tip of inner blue flame as measured with an accurate 1 percent full scale thermocouple pyrometer. A sheet of facial tissue conforming to UU-T-450 shall be suspended taut and horizontal 9-1/2 inches below the point of application of the flame to the cable specimen and at least 1/2 inch from the chamber floor, so that any material dripping from the cable specimen shall fall upon the tissue.

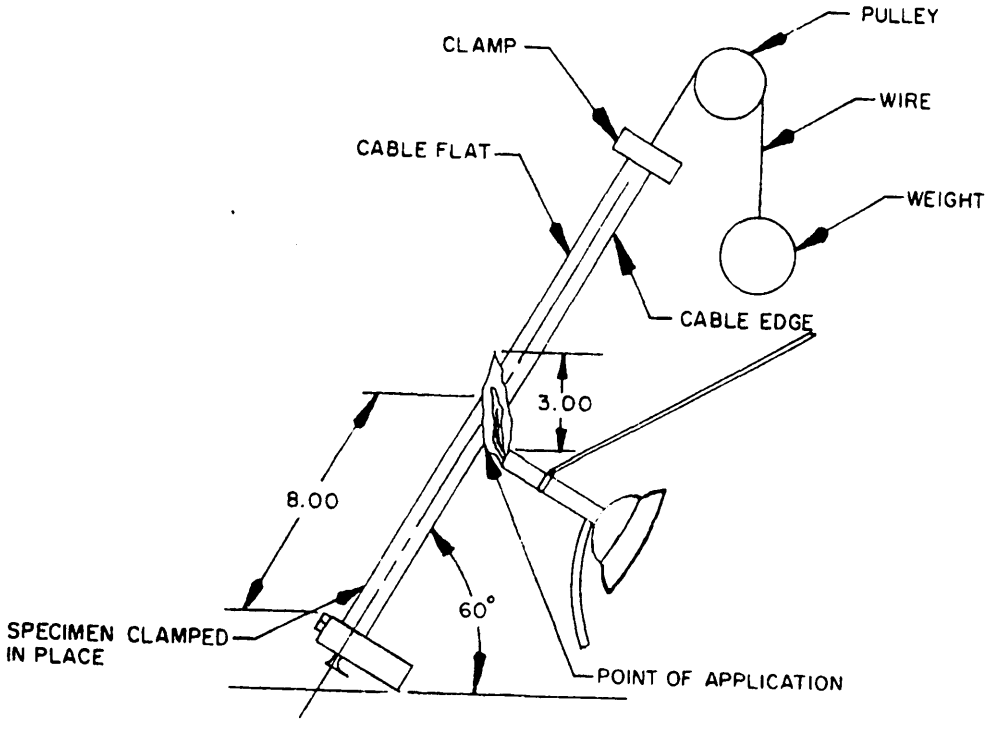


FIGURE 2. Flammability - test apparatus (shown without metal cabinet).

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4.6.4.2 Procedure. A 24-inch specimen of cable shall be marked at a distance of 8 inches from its lower end to indicate the point for flame application and shall be placed in the specified 60 degree position in the test chamber, with the flat of the cable in a horizontal position. The lower end of the specimen shall be clamped in position in the specimen holder and the upper end shall be passed over the pulley of the holder and held taut. With the burner held perpendicular to the specimen and at an angle of 30 degrees from the veritcal plane of the specimen (see figure 2), the tip of the inner blue flame shall be applied to the lower side of the cable at the test mark. The test flame shall be applied for 30 seconds for all sizes of cable then withdrawn immediately after 30 seconds. The distance of flame travel upward along the specimen from the test mark and the time of burning after removal of the test flame shall be recorded; also the presence or absence of flame in the facial tissue due to incendiary drip from the specimen. Charred holes or charred spots in the tissue shall be ignored in the absence of actual flame. Breaking of the cable specimens in sizes 24 and smaller shall not be considered a failure provided the requirements for flame travel limits, duration of flame, and absence of incendiary dripping are met.

4.6.5 Folding (see 3.8.1). One-quarter inch +.06 inch of insulation shall be removed from each end of a 24-inch ±1 inch specimen of cable and the specimen shall be folded 180° transversely as shown in figure 3 and placed between two smooth-surfaced flat metal plates. A load then shall be applied to the metal plates to achieve a pressure of 30 ±1 lb_f/in^2. The applied pressure shall be calculated considering only the triangular area included in the transverse fold. After 15 ±1 minutes, the specimen shall be unfolded and the same load shall be applied to the unfolded creased portion for an additional 15 ±1 minutes. This cycle (folding and unfolding) shall be repeated in the same direction on the same creased portion of the specimen and the specimen then shall be checked for conductor continuity. The specimen then shall be subjected to the dielectric withstanding voltage test (see 3.5.2).

4.6.6 Folding temperature test (see 3.8.2). One-quarter inch of insulation shall be removed from each end of a 24-in. specimen of cable and the specimen shall be folded 180° transversely as shown in figure 3 and placed between two smooth-surfaced flat metal plates. A load then shall be applied to the metal plates to achieve a pressure of 30 ±1 lb_f/in^2. The applied pressure shall be calculated considering only the triangular area included in the transverse fold. After 15 ±1 minutes, the pressure shall be released. The folded specimen then shall be conditioned for 4 hours ±15 minutes in an air-circulating oven at the following temperature.

Temperature rating (°C)	Test temperature (°C)
105	105±3
150	150±3
200	200±3

The velocity of air past the specimen (measured at room temperature) shall be between 100 and 200 feet per minute. The specimen then shall be transferred within 5 minutes to a cold chamber at -55 ±3°C. After 4 hours ±15 minutes, the specimen shall be allowed to return to room temperature and then checked for conductor continuity. The specimen then shall be subjected to the dielectric withstanding voltage test (see 3.5.2).

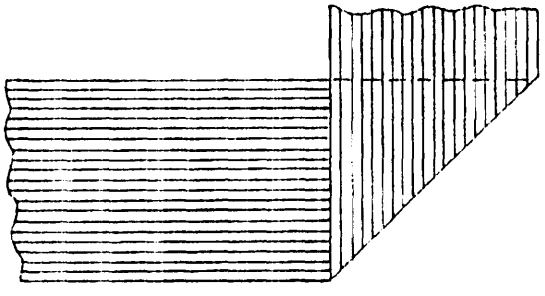


FIGURE 3. 180° transverse fold.

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4.6.7 High temperature aging (see 3.9).

4.6.7.1 Accelerated aging. 0.25 ±0.10 inch of insulation shall be removed from each end of an 18 ±5-inch cable specimen. The central portion of the cable then shall be bent at least halfway around a horizontally placed 0.25 in. diameter stainless steel mandrel. The stripped conductor ends shall be evenly clamped together between two stainless steel or aluminum strips and the cable shall be loaded with the following weight so that the portion of insulation between the conductors and mandrel is under compression while the conductors are under tension (see figure 4).

Equivalent round wire size (AWG)	Weight* (lb)
28 and 26	2 $^{+.3}_{-0}$
24 through 18	4 $^{+.3}_{-0}$

*Includes the weight of the clamp.

This specimen, so prepared on the mandrel, shall be conditioned in an air-circulating oven for 7 hours ±15 minutes at the following temperature.

Temperature rating (°C)	Test temperature (°C)
105	136
150	300
200	320

The velocity of air past the specimen (measured at room temperature) shall be between 100 and 200 ft per minute. After conditioning, the oven shall be shut off, the door opened, and the specimen allowed to cool in the oven for at least 1 hour. When cool, the specimen shall be freed from tension, the strips shall be removed and the specimen shall be removed from the mandrel. The central portion of the cable shall be bent at least halfway around the mandrel so that the portion which was outside during conditioning is now next to the mandrel. The cable shall be loaded with the applicable weight for 1 hour ±10 minutes. The cable shall then be removed from the mandrel and examined for cracking and separation of the insulation. The specimen shall then be subjected to the dielectric withstanding voltage test (see 3.5.2).

4.6.7.2 Life cycle. The cable shall be tested in accordance with 4.6.7.1 for the following time, at the following temperature, and using the following mandrel.

Temperature rating (°C)	Test temperature (°C) ±3°C	Time ±1 hour (hours)	Mandrel diameter (inch)
105	150	168	0.25
150	200	168	0.25
200	270	168	0.25

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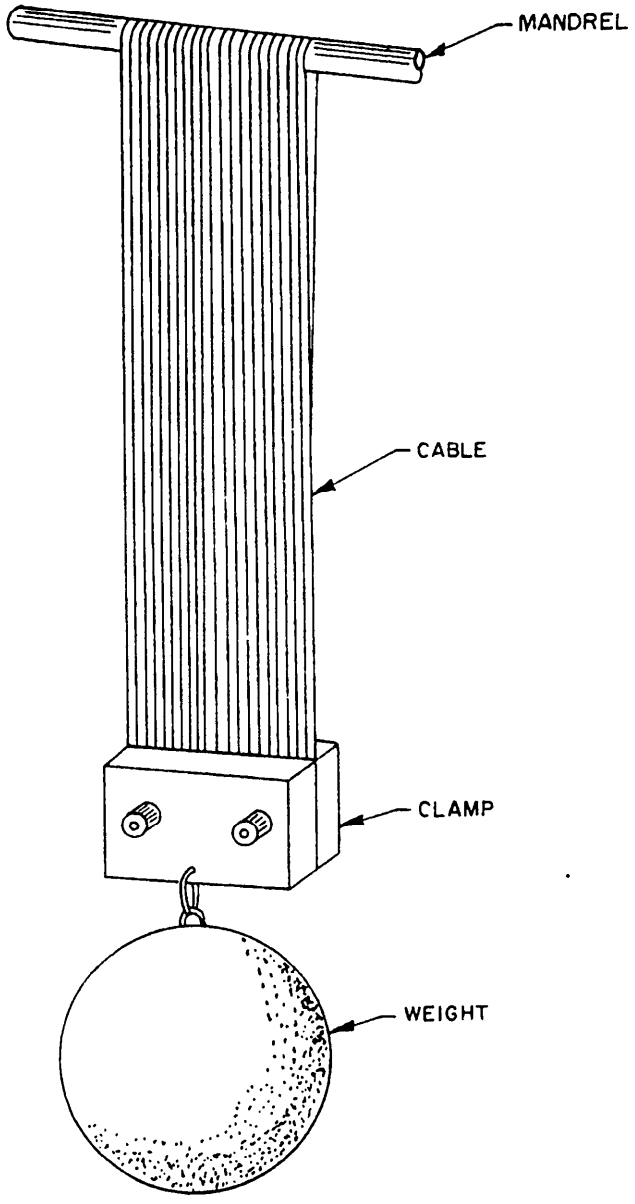
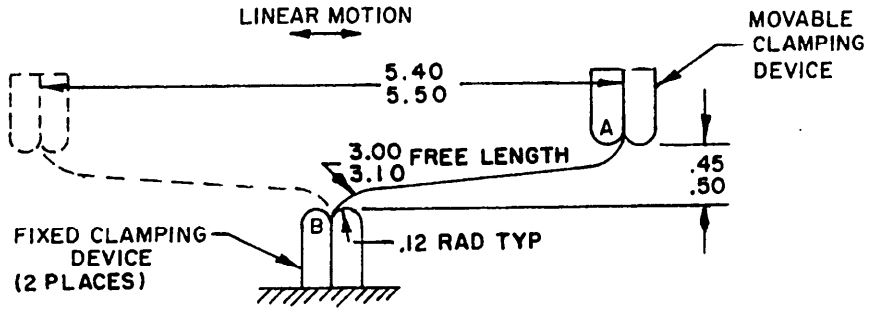


FIGURE 4. Accelerated aging and life cycle.

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NOTE: The free length between A and B shall be 3.00 to 3.05 inches.

FIGURE 5. Flex test fixture.

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4.6.8 Flexing endurance (see 3.10). A specimen of cable having a length of 6 inche (minimum) shall be installed in a fixture as shown in figure 5. All conductors of the test specimen shall be electrically connected in series during the test a monitoring current of 20 ±5 milliamperes shall be applied. A continuity tester capable of indicating a current interruption of 1 microsecond shall be employed to monitor the current. One end of the cable shall be held in the fixed clamping device. The other end shall b fastened in the movable clamping device, allowing a free length of 3.00 to 3.05 inches between clamping devices. The movable clamping device shall be driven in reciprocating linear motion for 5.4 to 5.5 inches from one extreme position to the other. One cycle shall consist of travel from one extreme to the other and return. The specimen shall b subjected to 30 cycles per minute for the number of cycles and at the temperature specified in the applicable specification sheet. A different cable specimen shall be used for flexing at each temperature.

4.6.9 Moisture resistance (see 3.11). A specimen of cable 28 to 40 inches long shall be tested in accordance with method 106 of MIL-STD-202 (steps 7a and 7b, polarization or loading voltage shall not be required). Following the tenth cycle and while the specimen is still in the humid environment, the insulation resistance shall be measured as specified in 4.6.2.3 with a potential of not less than 500 volts DC which shall be applied for a minimum of one minute between each conductor and its next adjacent conductors.

4.6.10 Low pressure (maximum rated temperature) (see 3.12). One specimen 14 ±1/16 inches long shall be prepared by cutting both ends approximately square. The sample shall be weighed on an analytical balance and the weight of the sample calculated to the nearest 0.001 gram. The uncoiled specimen shall be placed in a suitable open-rack (or equivalent) in a vacuum oven that has been preheated and stabilized, for not less than ten minutes, at the following temperatures:

Temperature rating (°C)	Test temperature °C ±3°C
105	105
150	150
200	200

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The oven shall then be allowed to restabilize at the above temperature. After temperature stabilization, the oven pressure shall be reduced to 2.0 torr ±.1, and maintained at these conditions for not less than 30 minutes. (A torr is nearly equal to the pressure of a column of mercury 1 millimeter high at 0°C at standard gravity.) On the completion of the 30-minute conditioning, the heat source shall be removed and the oven shall be allowed to cool to room ambient temperature. After room ambient temperature has been achieved, the chamber shall be brought to room ambient pressure. The specimen shall then be removed from the oven, and carfully examined. The insulation shrinkage, and the cable examination shall meet the requirements specified in 3.12. The specimens shall then be subjected to the dielectric withstanding voltage test specified in 4.6.2.2, followed by the insulation resistance test specified in 4.6.2.3, except that the testing shall be done in air between conductors only, and meet the requirements of 3.5.2 and 3.5.3, respectively.

4.6.11 Vacuum stability (see 3.13).

Test apparatus. The test apparatus shall consist of a vacuum system capable of 10^{-6} torr, a specimen heat-transfer block capable of maintaining 125°±5°C in a vacuum, a copper condensation block capable of being maintained at 25°±5°C in a vacuum, a microbalance (Mettler M5, or equivalent) capable of measuring to 1 µg, a desiccator, a humidity chamber at 50 percent relative humidity, a specimen holder, and a collector plate made from aluminum foil (see figure 6) (or equivalent).

Test procedure. Sufficient insulation to perform the test shall be removed from a cable specimen and wiped clean to ensure that contamination due to handling is not present during testing. The specimen holder and collector plate shall be wiped clean with acetone, weighed separately on the microbalance, and placed in the desiccator until ready for use. 0.1 to 0.8 g of insulation shall be weighed into the specimen holder and the specimen holder containing the insulation shall be placed in the humidity chamber for 24 hours at 50 percent relative humidity. After 24 hours, the specimen holder containing the insulation shall be weighed and then placed in the desiccator for 16 hours. After 16 hours, the specimen holder containing the insulation shall be removed from the desiccator, weighed, and placed within the heat-transfer block located in the vacuum chamber. At the same time the collector plate shall be attached to the copper condensation block located within the vacuum chamber. The location of the heat-transfer block must be such that all volatiles evolved from the insulation strike the collector plate. The vacuum chamber then shall be sealed and evacuated to 10^{-5} torr. The insulation then shall be heated to 125°±5°C (257° ±9°F) and maintained at that temperature for 24 hours. After 24 hours, heating shall be discontinued and the specimen holder allowed to cool for 1/2 hour to approximately 32°C (90°F). The vacuum chamber then shall be back-filled with dry nitrogen and the specimen holder containing the insulation and the collector plate shall be removed from the vacuum chamber and weighed or placed in a desiccator until they are weighed. The total weight loss (TWL) of the insulation and the volatile condensable materials (VCM) shall be calculated as follows:

$$TWL = 100\left[\frac{W_4 - W_5}{W_3 - W_1}\right] \qquad VCM = 100\left[\frac{W_6 - W_2}{W_3 - W_1}\right]$$

where: TWL = total weight loss of insulation (%)
VCM = volatile condensable materials (%)
W_1 = weight of specimen holder
W_2 = weight of collector plate
W_3 = weight of specimen holder containing insulation after removal from the humidity chamber
W_4 = weight of specimen holder containing insulation just prior to placing in the vacuum chamber
W_5 = weight of specimen holder containing insulation after removal from the vacuum chamber
W_6 = weight of collector plate after removal from the vacuum chamber.

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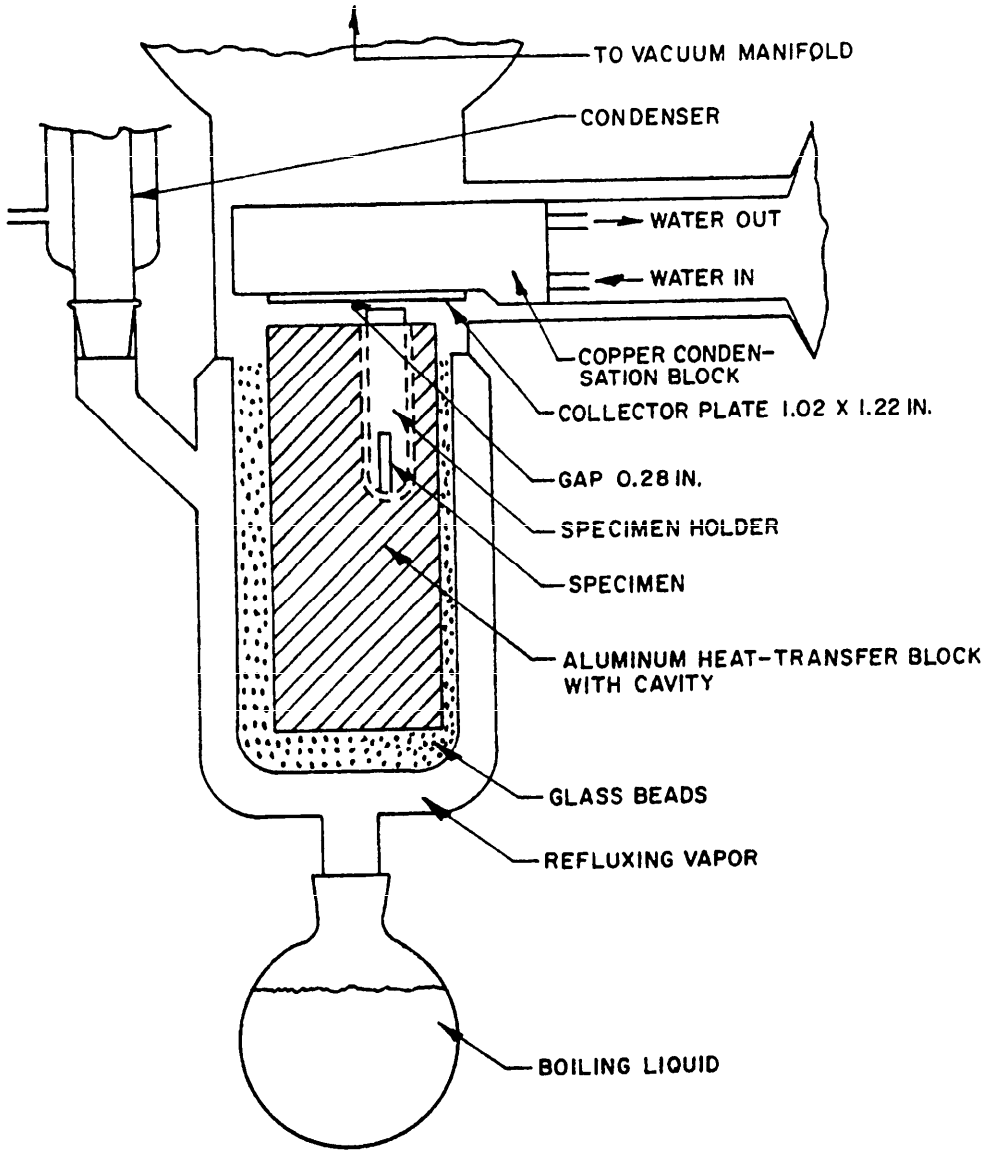


FIGURE 6. Schematic of vacuum stability test apparatus.

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4.6.12 Fluid immersion (see 3.14). 0.25 ±0.06 inch of insulation shall be removed from each end of four 24 inches ±1 specimen of cable. Each specimen shall then be weighed and one specimen shall then be immersed to within 6 inches of its ends in each of the fluids specified in table V for 20 ±0.5 hours at the temperature specified.

TABLE V. Test fluids and test temperatures.

Test fluid	Test temperature °C
TT-M-261 TT-I-735 MIL-T-5624 (JP-5)	20 to 22
MIL-H-5606 MIL-L-23699	68 to 72

During immersion the radius of bend of the specimen shall not be less than 0.5 inch. Upon removal from the fluids, the specimen shall remain for 1 hour in free air at room temperature. Each specimen then shall be weighed and compared with its initial weight. The specimens then shall be folded and unfolded twice in accordance with 3.8.1 and then subjected to the dielectric withstanding voltage test (3.5.2).

4.6.13 Blocking (see 3.15). One end of a specimen of cable of sufficient length to perform the test shall be affixed to a metal spool having a barrel diameter of at least 3.5 inches. The cable shall then be wound on the spool for at least three closely wound layers. The free end of the cable shall then be affixed to the spool or have a weight attached to it to prevent unwinding or loosening of the layers. The spooled specimen shall be conditioned in an air oven for 24 hours ±1 at the following test temperature:

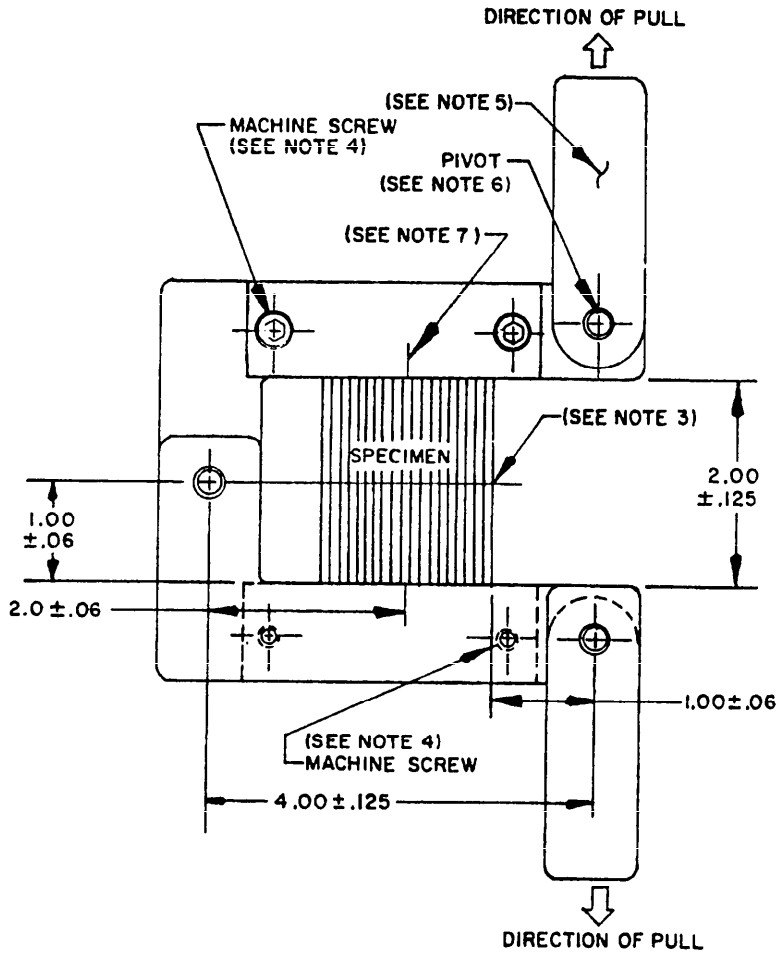
Temperature rating °C	Test temperature °C ±3°C
105	136
150	200
200	250

After conditioning, the spooled specimen shall be removed from the oven and allowed to cool to room temperature. After cooling, the specimen shall be unwound manually and examined for evidence of blocking.

4.6.14 Notch sensitivity (edge) (see 3.16). With the specimen centered on the scribe marks, the width of the specimen shall be 0.5 inch for cables having .050 or .100 inch conductor spacing and 0.6 inch for cables having 200 inch conductor spacing. A cable specimen at least 5 inches long shall be clamped in the specimen holder. The insulation shall be notched by a single-edged razor blade or equivalent, as shown on figure 7. The notch shall be made by attaching the razor blade and the outermost conductor to a continuity detector and cutting inward through the insulation until the continuity detector indicates that continuity has been made. The conductor must not be nicked. The holder shall be capable of clamping the cable specimen to prevent slipping of the specimen during the test. The specimen holder shall then be placed in the jaws of a tensile test machine and the machine shall be operated in the tensile mode at the jaw separation rate of 1 inch per minute. The test shall be discontinued when fracture occurs in the conductor adjacent to the notched edge of the insulation or when the tensile load exceeds 100 lbs per inch of cable width, whichever occurs first. The tensile load and any propagation of the notch in the insulation shall be recorded.

4.6.15 Twist (see 3.17). A cable specimen of sufficient length to allow 10 inches ±1 between clamps shall be mounted in the apparatus. The apparatus shall consist of two coaxially alined clamps. One clamp shall be capable of being rotated 180° ±3°

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INCHES	MM
.06	1.52
.125	3.18
1.00	25.4
2.00	5.08
4.00	10.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Insulation notched to conductor.
4. Machine screws 4 required.
5. Place into jaws of tensile test machine.
6. Pivot pins 3 required.
7. Scribe mark two places.

FIGURE 7. Notch sensitivity (EDGE) specimen holder.

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in each direction about the longitudial axis of the specimen. The other clamp shall be incapable of rotational motion but shall be capable of motion along the longitudinal axis of the specimen so as to maintain a specimen held between the clamps at a tension load of 2 lbs ±.1 per inch of specimen width during the test. The specimen shall then be rotated 100 cycles at a rate of 12 cycles per minute. One cycle shall consist of rotating the specimen 180° ±3° in one direction from the initial position, returning to the initial position, rotating 180° ±3° in the opposite direction, and returning to the initial position. During the test, the specimen shall be observed and if separation of insulation between conductors occurs prior to 100 cycles the test shall be discontinued. After 100 cycles, the specimen shall then be removed from the clamps and examined visually for cracking or separation of the insulation.

4.6.16 Shrinkage (see 3.18). A 12-inch ±.5 specimen of cable shall be cut so that the insulation and conductor are flush at both ends. The specimen then shall be conditioned in an air-circulating oven for 6 hours at the following temperature:

Temperature rating (°C)	Test temperature (°C) ±3°C
105	136
150	200
200	250

The velocity of air past the specimen (measured at room temperature) shall be between 100 and 200 ft per minute. After conditioning, the specimen shall be removed from the oven and allowed to cool to room temperature. The shrinkage of the insulation then shall be measured as the greatest distance which the insulation has receded from either end of the conductors; that is, the measurement obtained at the end showing the greater shrinkage shall be considered the shrinkage of the specimen.

4.6.17 Strippability (see 3.19). The specimen of completed cable shall have the insulation removed from approximately 1/4 inch at each end using a cold blade stripper (MSFC Drawing Number 95M20109, or equivalent). When stripped, the specimen shall meet the requirements of 3.19.

4.6.18 Solderability (see 3.20). Each conductor shall be subjected to method 208C of MIL-STD-202, with the following exceptions with reference to preparation of cable as follows:

a. Straighten and cut wire into convenient lengths (2 inches minimum).
b. Degrease by immersion in either acetone or trichloroethane for two minutes
c. Clean in fluoroboric acid (HBF_4), 10 percent water solution, for 5 minutes with agitation. Use caution in handling.
d. Rinse acid off as follows:
   1. Two cold water rinses.
   2. Two isopropyl alcohol rinses.
e. Immerse in flux (MIL-F-14256, type RMA).
f. Dip in molten solder for 5 seconds at 230° ±5°C (446° ±9°F).
g. To remove or dissolve the residual type RMA flux, wash or rinse in isopropyl alcohol.

NOTE: All chemicals shall be of commercial grade or better. Fresh solvents shall be used as often as is necessary to preclude contamination.

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4.6.19 Durability of identification.(see 3.22). The durability of product identification shall be evaluated at room temperature as follows:

Test apparatus. A General Electric Repeated-Scrape Abrasion Tester, Model 158L238G1, or equivalent shall be used. The instrument shall be capable of rubbing a small cylindrical steel mandrel 0.025 to 0.028 inch in diameter repeatedly over the upper surface of the cable in such position that the longitudinal axes of the steel mandrel and specimen are at right angles to each other with surfaces in contact. A weight affixed to a fixture above the rubbing mandrel shall control the force exerted normal to the surface of the insulation. A motor-driven, reciprocating-cam mechanism shall be used to deliver an accurate number of abrading strokes in a direction parallel to the longitudinal axis of the specimen. The number of strokes shall be measured by a counter. The length of the stroke shall be 0.375 inch (minimum) and the frequency of the stroke shall be 60 cycles (120 strokes) per minute.

Test procedure. A test specimen shall be prepared by removing from the cable at least a 4-inch length of single insulated conductor containing identification. The specimen shall be mounted in the specimen holder and 150 grams shall be applied through the steel mandrel to the marked surface of the specimen. The counter shall be set at zero and the drive motor started. The specimen shall be subjected to the specified number of cycles as stated in 3.22 and shall be examined. If a continuous line of identification marking has been erased or obliterated by the mandrel, the specimen shall be considered as having failed.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with the requirements of MIL-C-12000.

5.2 Reels and spools. Unless otherwise specified, spool and reel barrels shall be not less than 3 inches in diameter, and the flanges shall be not greater than 18 inches in diameter.

5.3 Continuous lengths. Unless otherwise specified in the contract the minimum acceptable finished cable lengths shall conform to table VI.

TABLE VI. Finished cable lengths.

Cable width	Minimum percentage of footage in shipment with lengths greater than:			
in.	100 ft	50 ft	25 ft	10 ft
0.5	50	80	--	100
1.0	--	50	80	100
2.0	--	50	80	100

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6. NOTES

6.1 Intended use. The electric multiconductor cable covered by this specification is intended for use with connectors of the Standard Electronic Modules and in any application where its performance characteristics are required. It is suitable for installation in aerospace and military electrical systems or for general purpose use within the limitations of the applicable performance requirements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number and date of the applicable specification sheet (see 3.1).
c. Applicable military part number (see 3.1).
d. Total number of feet and acceptable minimum continuous length.
e. Color if other than natural is required (see 3.4.2.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Quality Products List whether or not such products have actually been so listed by that date. The attention of the contactors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the Qualified Products List is U.S. Army Electronics Command, Fort Monmouth, NJ 07703, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" to DESC-EQ, 1507 Wilmington Pike, Dayton, OH 45444.

6.3.1 Contract supplied cable. It is understood that flat multiconductor cable supplied under contract shall be identical in every respect to the sample tested and found satisfactory except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

6.4 Definitions.

6.4.1 Blocking. Blocking is the transfer of insulation from one layer of cable to another.

6.4.2 Conductor. A conductor is a single conductive path in a conductive pattern.

6.4.3 Conductor thickness. The conductor thickness is the thickness of the conductor including all metallic coatings.

6.4.4 Conductor width. The conductor width is the observable width of the conductor at any point chosen at random along the cable viewed from vertically above.

6.4.5 Current-carrying capacity. The current carrying capacity is the maximum current that can be continuously carried through a conductor under specified conditions.

6.4.6 Design width of conductor. The design width of a conductor is the width of a conductor specified as nominal and agreed on between the purchaser and manufacturer.

6.4.7 Dielectric strength. The dielectric strength is the maximum voltage that a dielectric material can withstand, under specified conditions, without rupturing; usually expressed as volts/unit thickness.

6.4.8 Flat cable. Flat cable is any cable with two or more parallel conductors in the same plane enclosed by an insulating material.

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6.4.9 Insulation resistance. Insulation resistance is the electrical resistance of the insulating material, determined under specified conditions, between any pair of conductors.

6.4.10 Margin. The margin is the distance between the edge of a flat cable and the near edge of the nearest conductor.

6.4.11 Solderability. Solderability is the property of a metal to be wetted by solder.

6.4.12 Twist. A twist is a deformation of a rectangular sheet such that one of the corners is not in the plane containing the other three corners.

6.4.13 Wetting. Wetting is the free flow and spreading of solder on conductive paths and terminals to form an adherent bond.

Custodians:
Army - CR
Navy - AS
Air Force - 11

Review activities:
Army - AR, AV, MI
Navy - EC, OS
Air Force - 17, 99
DLA - IS, ES
NSA
NAS
NASA - MSFC

User activities:
Army - ME, AT
Navy - SH
Air Force - 71, 99

Preparing activity:
Army - CR

Agent:
DLA - ES

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